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Project Title:	Quantitative Studies of Cool Flame Transitions at Radiation/Stretch Extinction Using Counterflow Flames		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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	Cool flames have significant implications for the control of engine knocking, ground and space fire safety, and combustion emissions. The recent International Space Station (ISS) combustion science experiments on flame extinguishment discovered the existence of cool flames for both large single droplets and droplet arrays after radiative extinction. Numerical simulations of droplet combustion also provide an explanation for the transition of hot flame to cool flame after radiative extinction. However, it remains unclear at what conditions the hot flames transfer to cool flames and how the cool flame flammable region is related to hot flame flammability limit and extinction limit. It is also not clear how oxygen concentration, dilution, pressure, diffusion, and convection transport processes affect the cool flame transition. More importantly, due to limits of space and payload, the ISS experiments did not provide detailed speciation and temperature distributions of cool flame structures for quantitative validation of kinetics and radiation		

Task Description:	model. By using the Flame Extinguishment (FLEX) experimental data from the Physical Sciences Informatics (PSI) system of combustion science, we propose to conduct new simulations and ground-based experiments using ozone-sensitized counterflow flames to * develop a new diagram of the flammability limits of diffusional cool flames affected by radiation, diffusion, and convection ; * bridge the knowledge gap between the flammability limits and extinction limits of hot flames and cool flames ; * predict and answer the question at what conditions the transition from hot flame extinction to cool flames is possible and verify the phenomena observed on the ISS flame extinguishment experiments ; * obtain new, quantitative data of extinction limit, flame temperature, and species distribution for the validation of kinetic and radiation models involving cool flames ; * extend the cool flame experiments from the original ISS droplet experiments to a more general ground-based combustion system involving radiation, diffusion, convection, and low-temperature chemistry while maintaining a one-dimensional geometry. The proposed new ground-based experiments and data analysis are complimentary to and not directly linked with the science objectives from the original ISS combustion science investigation, and will significantly advance the understanding of near-limit cool flame dynamics and contribute to the future control of engine knocking, ground and space fire safety, and combustion emissions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The cool flame extinction limit measurements for C7 to C12 n-alkanes showed that cool flame extinction experiment is a new platform to study the low temperature reactivity of transportation fuels and can provide crucial experimental targets for rapid screening of alternative fuels. The results also showed cool flame extinction limits is a critical validation target of low temperature chemistry because of the lack of information of the cool flame heat release rates in other experimental systems. The results also provide a new flammability limit diagram for both hot and cool premixed flames. The results showed that both the exhaust gas recirculation and the engine pressure dramatically affect the burning limits of cool flames. This is critical to the lean burn engine design. The counterflow cool flame extinction provides a good validation target and explanation for the NASA droplet cool flame extinction limit.
Task Progress:	We made two progresses in year 1 research. In the first research, the low-temperature oxidation of hydrocarbon fuels has received increasing attention as advanced engines seek to operate in less conventional combustion regimes. Large n-alkanes are a notable component of many real transportation fuels and possess strong reactivity in this important low-temperature range. These n-alkanes have been studied extensively in various canonical kinetic experiments but seldom in systems with strong coupling between low-temperature chemistry, transport, and heat release. To address this issue, the present study investigates self-sustaining n-alkane cool diffusion flames in a counterflow burner. The extinction limits of both hot diffusion flames and cool diffusion flames are measured at atmospheric pressure for a range of n-alkanes from n-heptane to n-tetradecane. It is observed that while these fuels behave similarly for hot flames, the larger n-alkanes are substantially more reactive in the low-temperature cool flame regime. Moreover, ozone addition strongly enhances the low-temperature chemistry to the point where the differences in fuel reactivity are nearly suppressed. The experimental measurements are compared with numerical simulations employing both detailed and reduced chemical kinetic models of various sizes. Although the different kinetic models adequately predict the extinction limits of the hot flames, a large scatter is present in the model results for cool flames, and a general overprediction of the measured cool flame extinction limit is observed for all of the fuels studied. This implies that the cool flame heat release is not well agreed upon by the current chemical kinetic models, despite their capability to reproduce many homogeneous reactor experiments at low temperatures. Furthermore, it is observed that the cool flame heat release is spread over a substantial number of reactions involving large molecules, a trait that makes it particularly difficult to create reduced kinetic models that can accurately describe cool flame behavior. The results of this study suggest that the cool flame platform can provide crucial validation of the coupling between chemistry, transport, and heat release in flames at low temperatures. In the second research, the flame speeds and propagation limits of premixed cool flames at elevated pressures are numerically modeled using dimethyl ether mixtures. The primary focus is paid on the effects of pressure, mixture dilution, computation domain, and heat loss on cool flame propagation. The results showed that cool flames exist on both fuel lean and fuel rich sides and dramatically extend the lean and rich flammability limits of conventional hot flames. There exist three different flame regimes: the hot flames, lean and rich cool flames, and double flames. A new flame flammability diagram including both cool flames and hot flames at elevated pressures is obtained. The results show that pressure significantly changes cool flame propagation and burning limits. It is found that the increase of pressure affects the propagation speeds of lean and rich cool flames differently due to the negative temperature coefficient effect. On the lean side, the increase of pressure accelerates the cool flame chemistry and shifts the transition limit of cool flame to hot flame to a lower equivalence ratio. At lower pressure, there is an extinction transition from hot flame to cool flame. However, above a critical pressure, the hot flame directly transfers to a cool flame without hot flame extinction. Moreover, increases in dilution reduce the heat release of the hot flame and promote cool flame formation. Furthermore, the results show that a smaller downstream computation domain and higher heat loss also extend the cool flame transition limit and promote cool flame formation.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Ju Y. "On the propagation limits and speeds of premixed cool flames at elevated pressures." Combustion and Flame. 2017 Apr;178:61-9. https://doi.org/10.1016/j.combustflame.2017.01.006 , Apr-2017
Articles in Peer-reviewed Journals	Reuter CB, Lee M, Won SH, Ju Y. "Study of the low-temperature reactivity of large n-alkanes through cool diffusion flame extinction." Combustion and Flame. 2017 May;179:23-32. https://doi.org/10.1016/j.combustflame.2017.01.028 , May-2017